

Industrial Internship Report on " Employee Payroll Management System"

Prepared by
Rajeyo Haldar

Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt. Ltd. (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to complete the project along with the report within 6 weeks.

My project was "Employee Payroll Management System", a Core Java application developed to automate employee record management and payroll processing. The system allows users to add, update, search and delete employee records, automatically calculates salary components such as HRA, DA, Bonus and Tax, generates employee payslips, and stores employee data using file handling. The project was designed using Object-Oriented Programming principles to provide an efficient, accurate and user-friendly payroll management solution.

This internship gave me a very good opportunity to gain exposure to industrial problem-solving and to design and implement a practical software solution. It was an overall great learning experience that enhanced my Core Java programming, problem-solving and software development skills.

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1 Preface

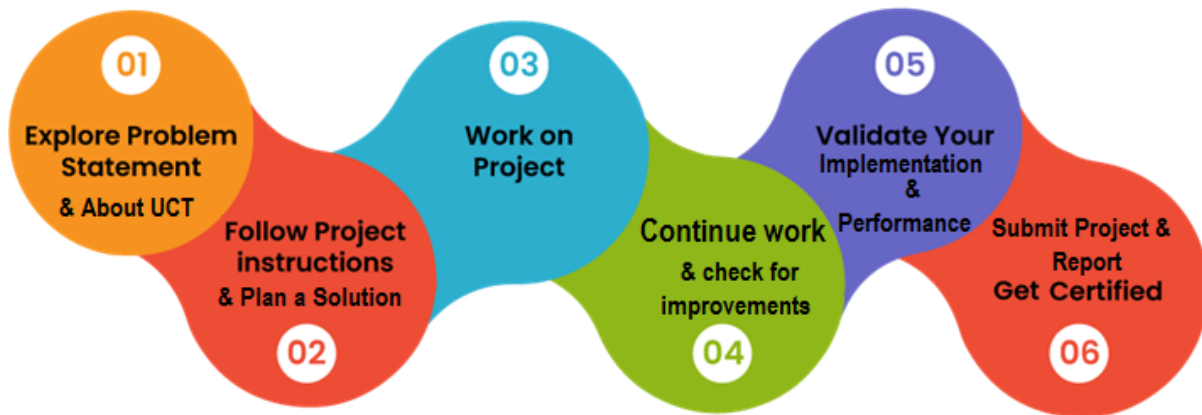
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



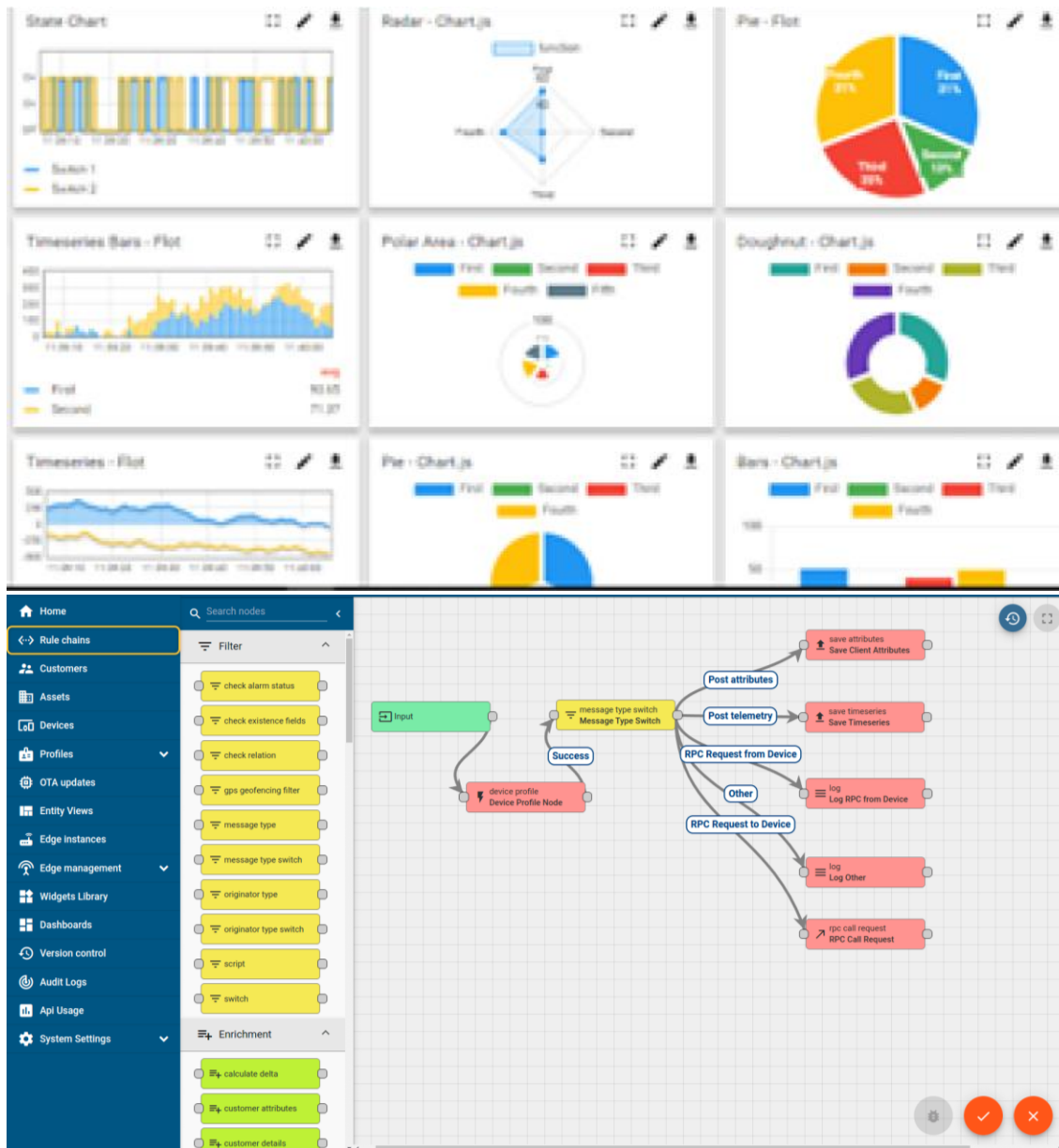
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (**FACTORY** **WATCH**)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



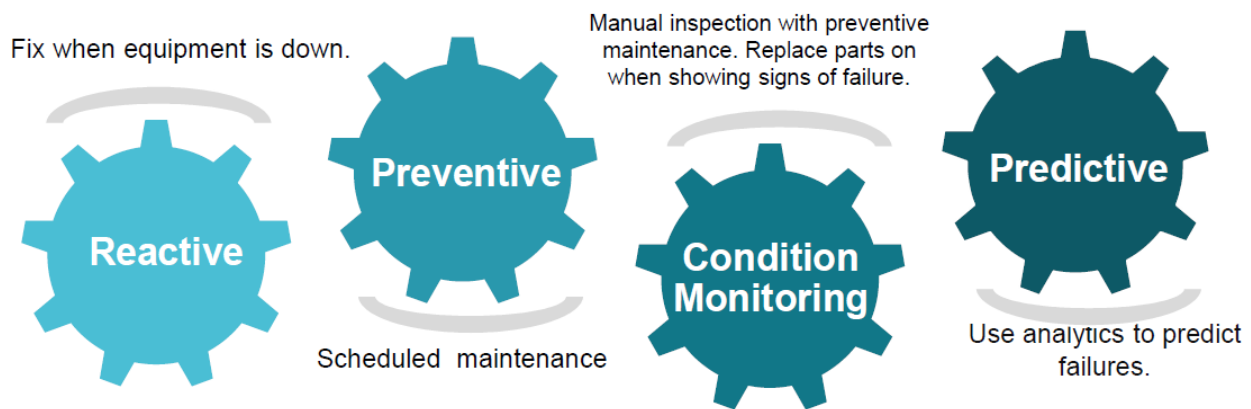


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

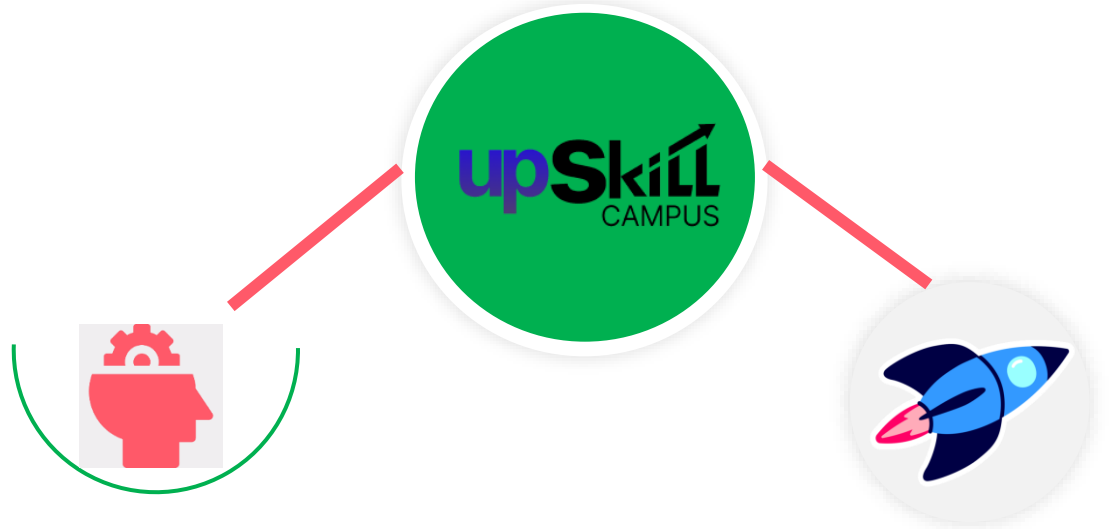
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

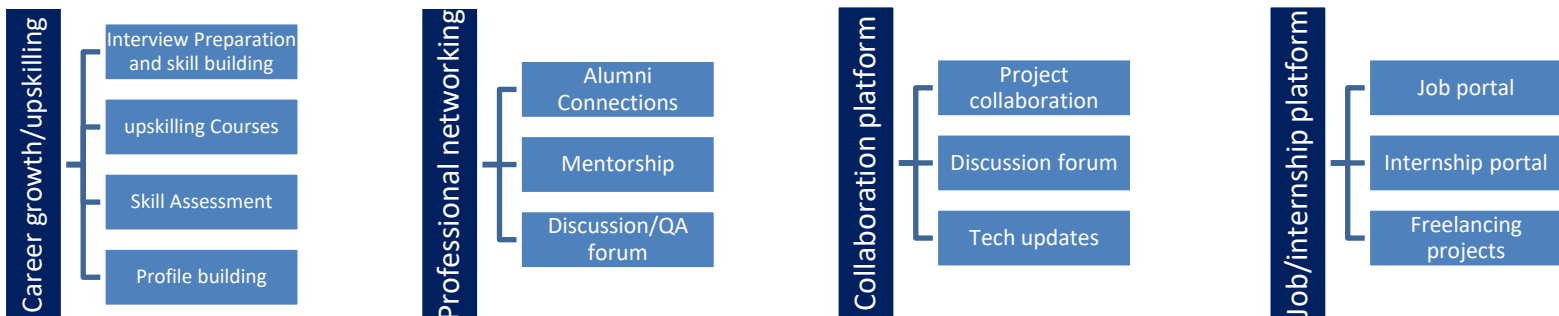
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Oracle Java Documentation. <https://docs.oracle.com/en/java/>
- [2] Oracle Java Tutorials (Object-Oriented Programming & File Handling).
<https://docs.oracle.com/javase/tutorial/>
- [3] Upskill Campus & UniConverge Technologies (UCT) Industrial Internship Guidelines and Project Documentation.

2.6 Glossary

Terms	Acronym
Employee Payroll Management System	EPMS
Human Resource Allowance	HRA
Dearness Allowance	DA
Object-Oriented Programming	OOP
Java Development Kit	JDK

3 Problem Statement

The objective of this project is to develop an **Employee Payroll Management System** using Core Java that simplifies employee record management and automates payroll processing. In many small and medium-sized organizations, employee salary calculations and record maintenance are performed manually or using spreadsheets, which can lead to calculation errors, duplicate records, data inconsistency, and increased processing time.

The proposed system provides a user-friendly solution to manage employee information, calculate salary components such as Basic Salary, HRA, DA, Bonus, Tax, and Net Salary automatically, and generate employee payslips accurately. It also allows users to add, search, update, and delete employee records while maintaining data efficiently. The project aims to improve accuracy, reduce manual effort, and demonstrate the practical implementation of Core Java concepts such as Object-Oriented Programming, Collections, File Handling, and Exception Handling.

4 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

Many organizations, especially small businesses, manage employee records and payroll manually using paper registers or spreadsheet applications such as Microsoft Excel. Although these methods are simple to use, they have several limitations:

- Manual salary calculations are time-consuming and error-prone.
- Employee records are difficult to organize and maintain.
- Updating employee information requires additional effort.
- Generating payslips manually takes more time.
- There is a higher risk of data duplication, inconsistency, and loss.
- Manual systems are not suitable for handling a large number of employees efficiently.

What is your proposed solution?

The proposed solution is a **Core Java-based Employee Payroll Management System** that automates employee record management and payroll processing. The application enables users to add, search, update, and delete employee records through a menu-driven interface. It automatically calculates salary components such as HRA, DA, Bonus, Gross Salary, Tax, and Net Salary, generates employee payslips, and stores employee information using file handling for future access. The system is developed using Object-Oriented Programming (OOP) concepts to ensure modularity, maintainability, and reliability.

What value addition are you planning?

The proposed system offers the following improvements over existing manual methods:

- Reduces manual effort and calculation errors.
- Automates payroll processing with accurate salary calculations.
- Provides quick access to employee records through search functionality.
- Generates employee payslips instantly.

- Maintains employee data efficiently using file handling.
- Improves data accuracy, consistency, and reliability.
- Demonstrates practical implementation of Core Java, OOP, Collections, and Exception Handling concepts, making it suitable for real-world applications and future enhancements.

4.1 Code submission (Github link) :-

<https://github.com/rajeyo316/upskillcampus>

4.2 Report submission (Github link) :_ first make placeholder, copy the link.

5 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

5.1 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

6.1 Test Plan/ Test Cases

6.2 Test Procedure

6.3 Performance Outcome

7 My learnings

You should provide summary of your overall learning and how it would help you in your career growth.

8 Future work scope

You can put some ideas that you could not work due to time limitation but can be taken in future.